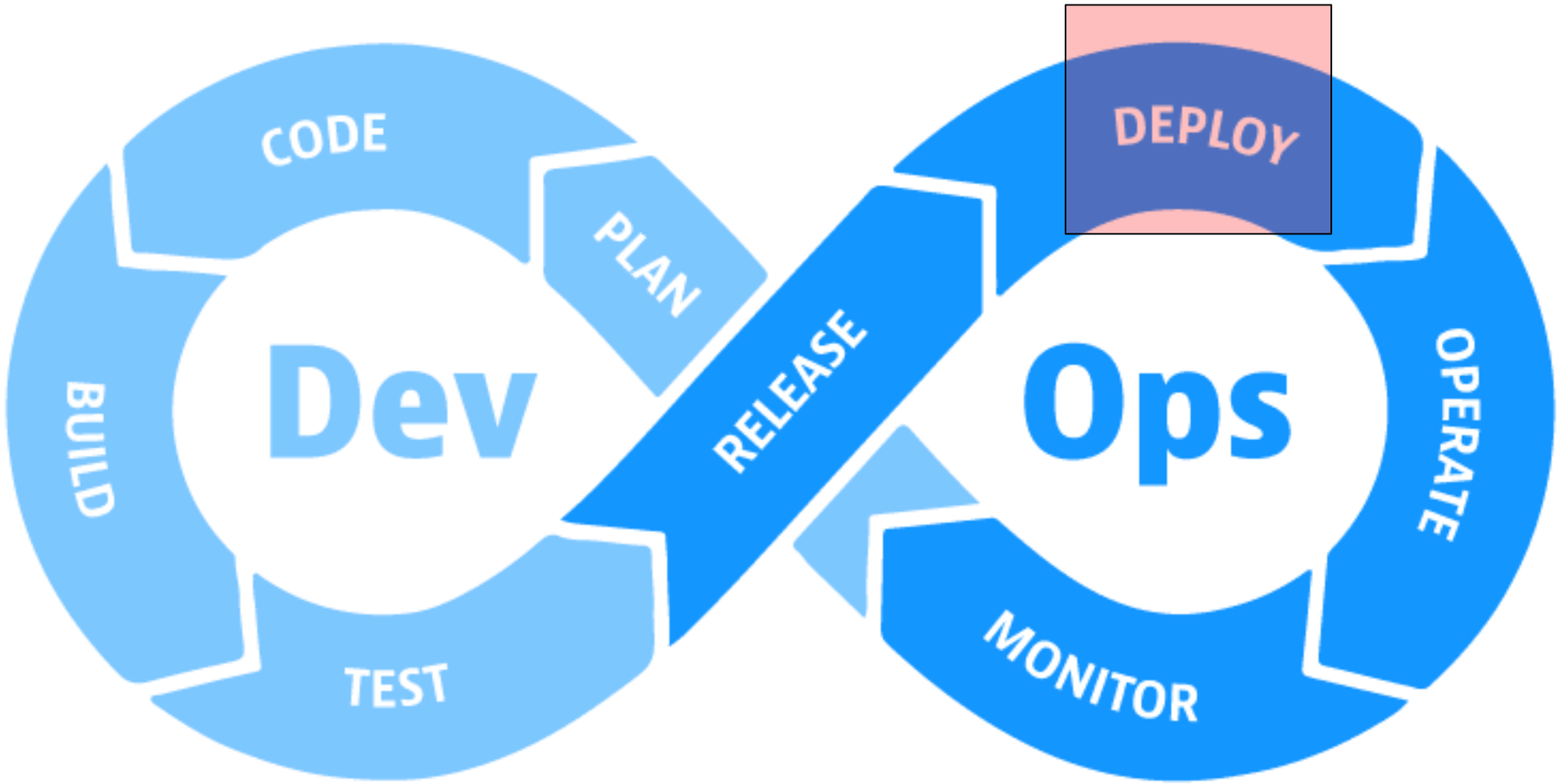
Developer Operations (devops)

Deployment – Helm + Stages

Deployment


CODE
PLAN
BUILD
Dev
TEST
RELEASE
DEPLOY
Ops
OPERATE
MONITOR

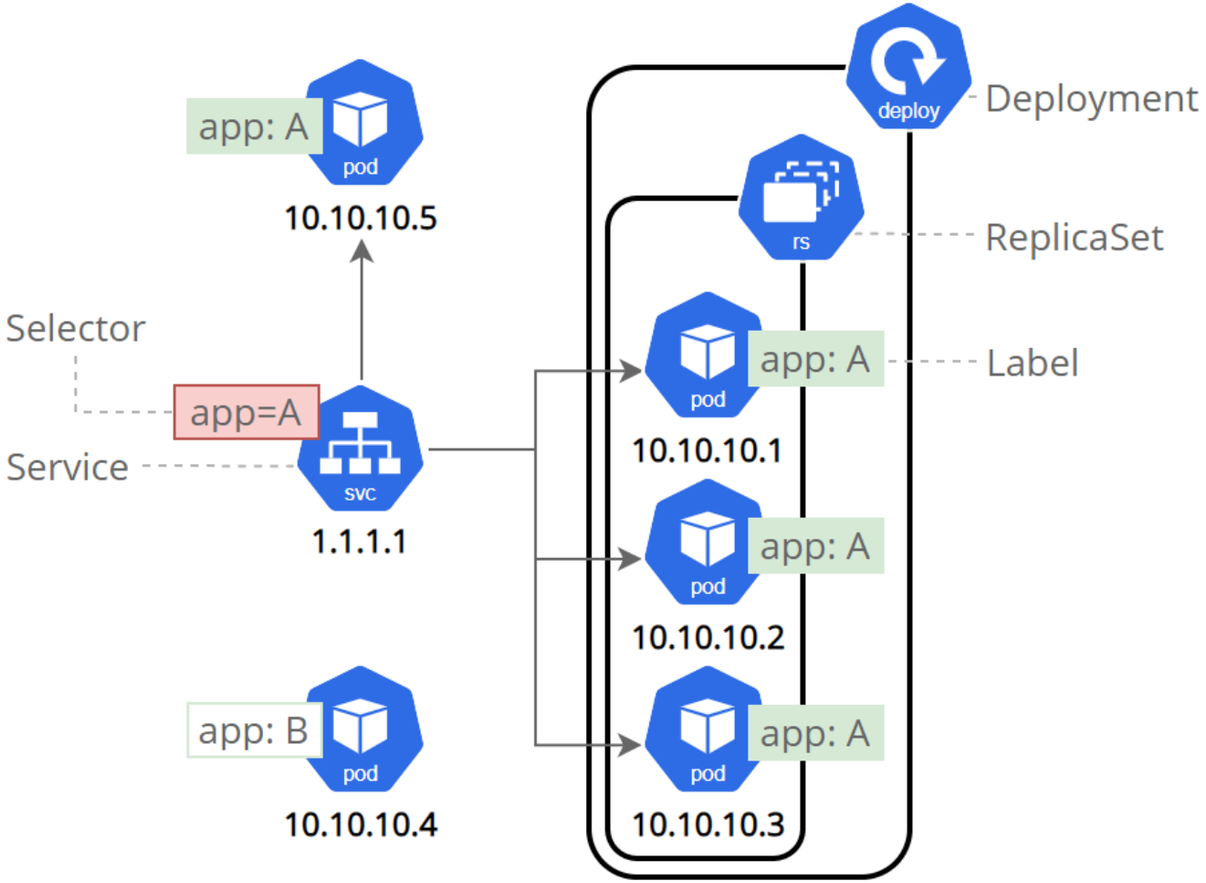
What‘s it all about today?

- Kubernetes Recap
- Challenges with Kubernetes manifest management
- Kubernetes packaging (e.g. Helm)
- Deployment Stages

Recap, Elements within Kubernetes

Objects, necessary for deploying an app

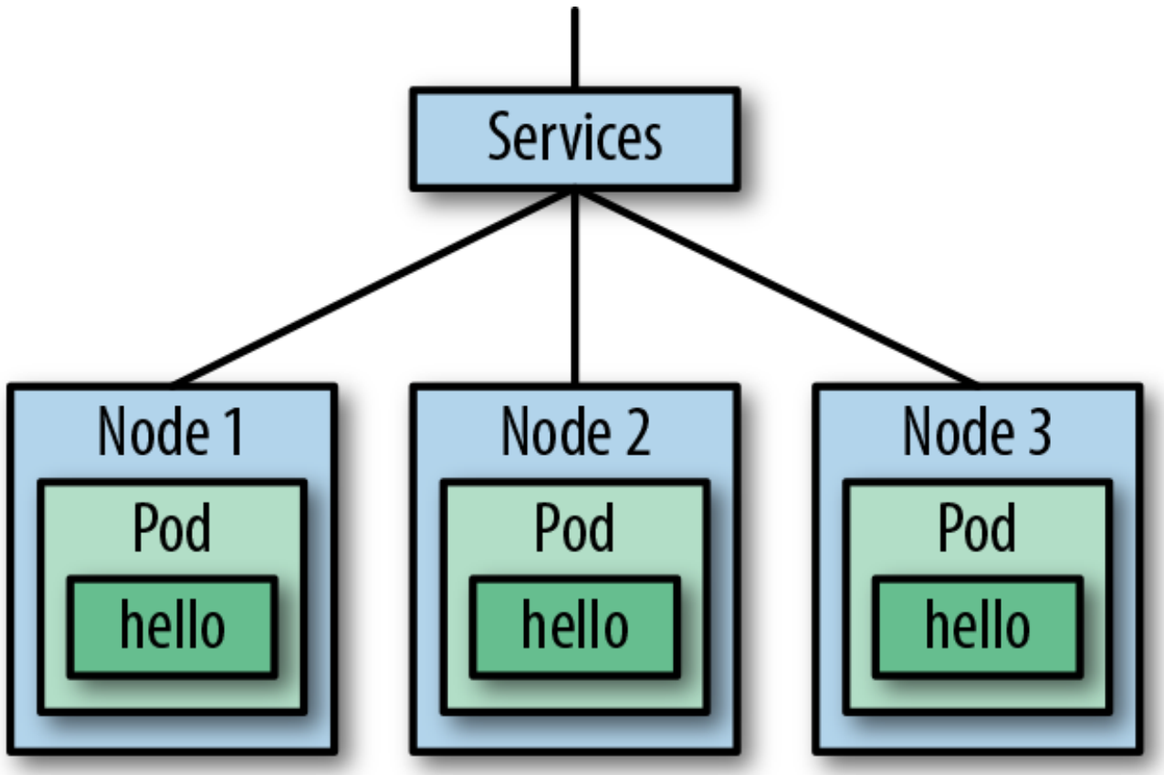
- **Deployment**: Description of target to deploy service
- **ReplicaSet**: Ensuring that pods are running
- **Pods**: Single set of running container
- **Service**: Address + Portmapping to a set to pods defined by label



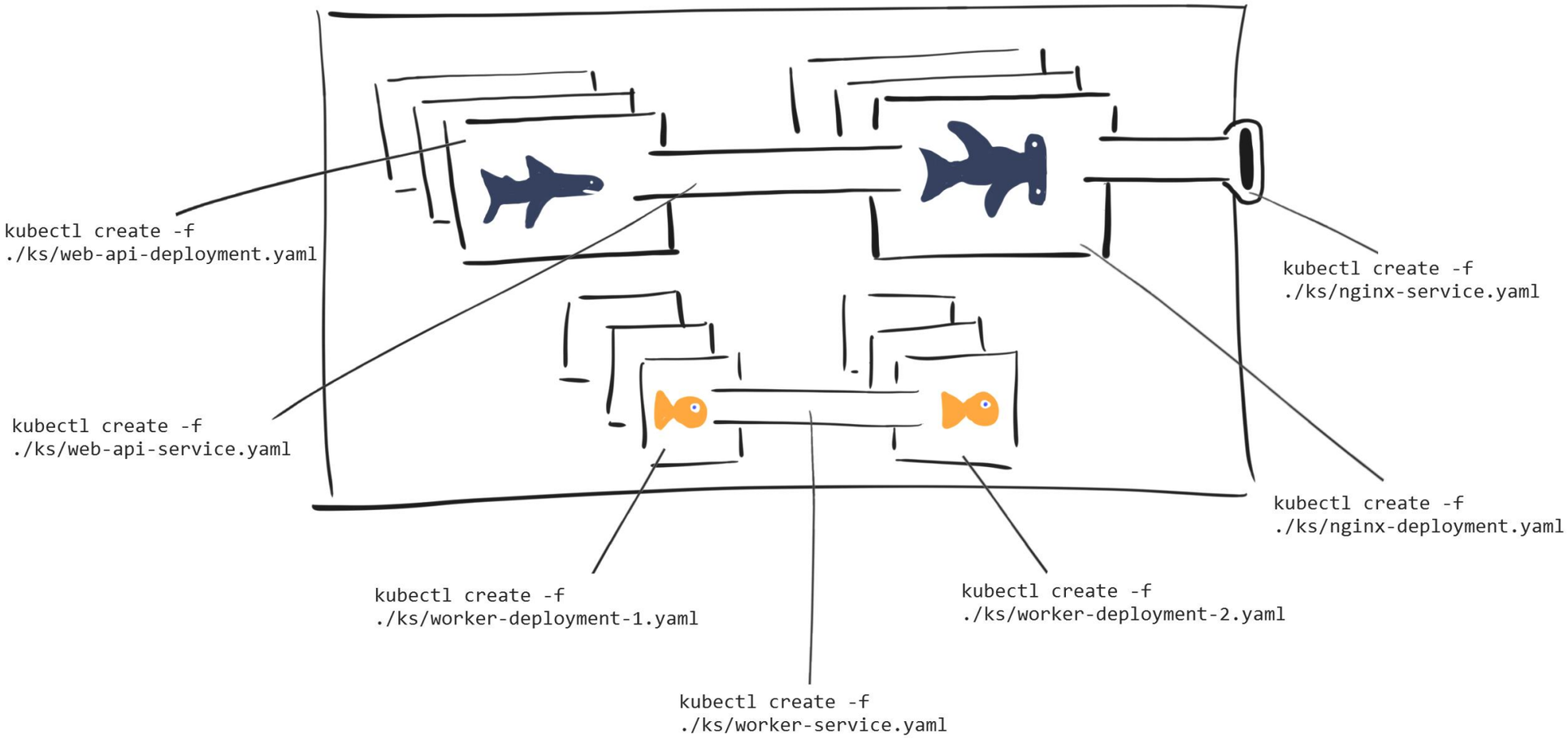
Service

- Traffic is bound to ports and containers
- Containers are ephemeral and might be deleted/created/etc.
- Multiple pods might represent the same service

Abstract is needed: a Service defines an open port forwarding traffic to all pods with a defined label



Kubectl, Problems?



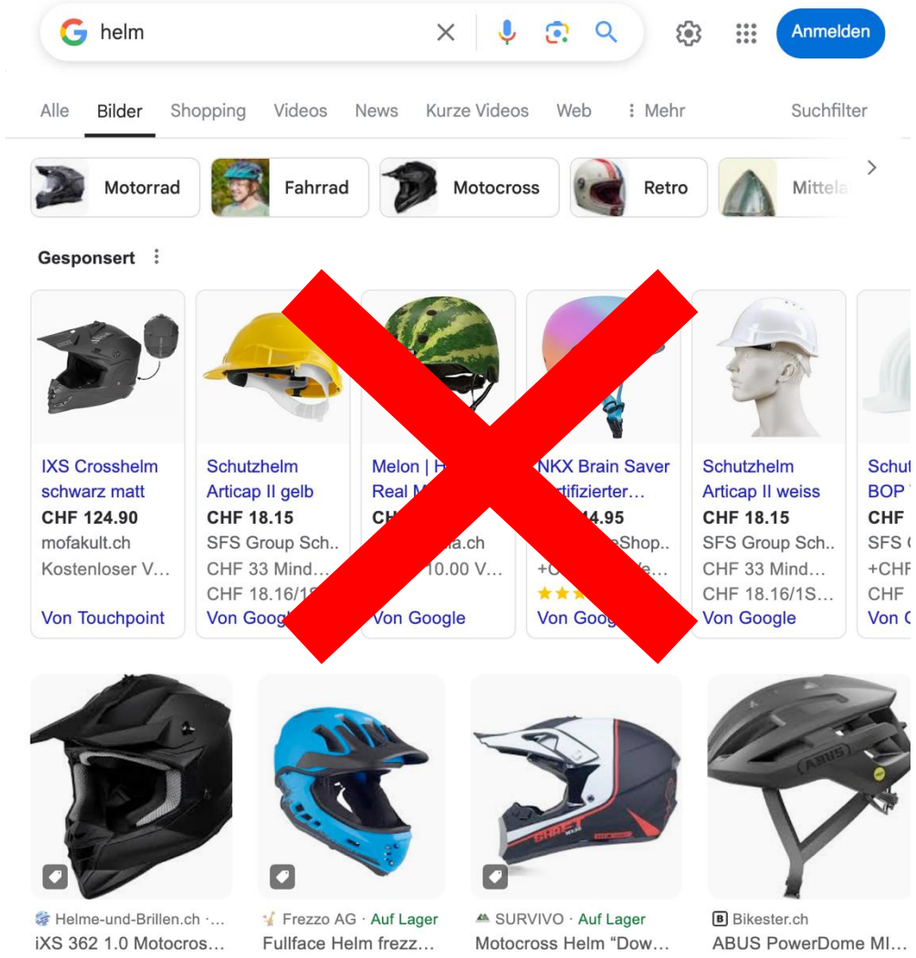
Common Antipatterns within Kubernetes Deployment

1. Putting configuration near the image generation
2. Not using Helm / Kustomize
3. Deploying things in specific order
4. Pulling latest images
5. Deploying due to killing pods

Project Intro

Origin

- Naming
  - Kubernetes: greek for "helmsman" or "pilot“
  - Helm: greek again; „steering mechanism of ship“
- Started 2015 by Deis (now Microsoft Azure)
- Graduated CNCF project (2020)
- De-facto standard for packaging of 3rd party Kubernetes apps



https://helm.sh/

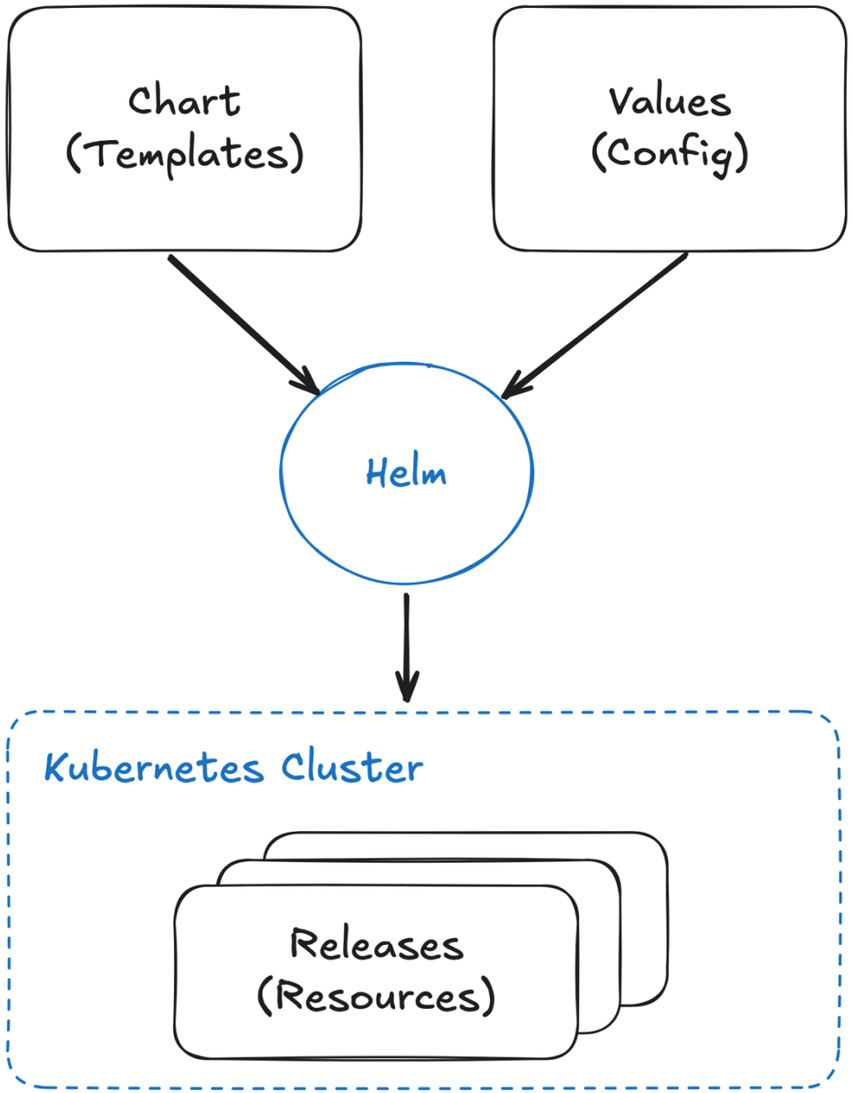
Helm - Concepts

Chart: Helm package, containing all resources to run an application inside Kubernetes

Values: Configration for one released chart.

Release: Instance of installed chart with provided configuraiton. Can occur multiple times inside a cluster in different namespaces. Each release has one defined release name.

„Kubernetes resource templates are merged with Values to create resulting Release“



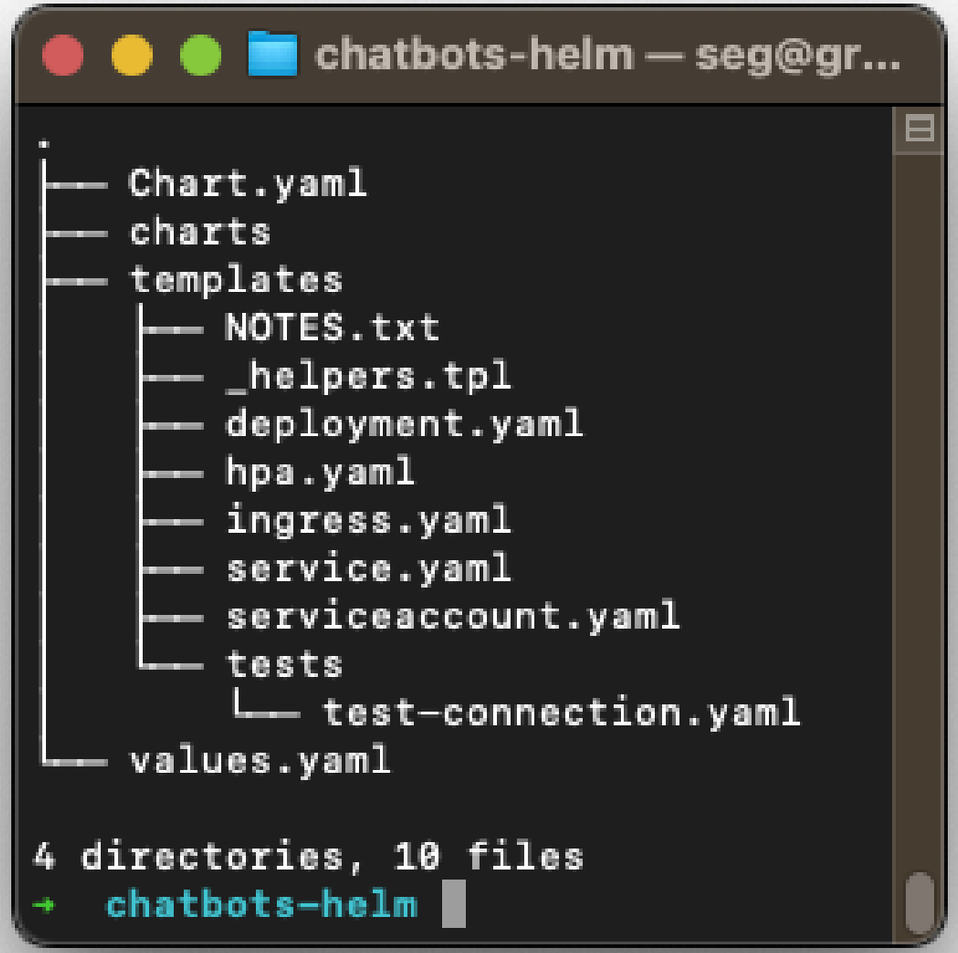
Source: Krusche & Company GmbH

Structure and Usage

Defined folder structure

- templates: K8s-resources to be deployed
  - Should contain entire architecture
  - Different values should be replaced by references
- values.yaml:
  - Replacing values in templates/resources
- Chart.yaml:
  - Metainformation about chart
  - Reference to subcharts

- CLI-Tool
  - Takes folder structure or packaged chart as input



Packaging

- Helm Charts can be packaged and distributed
  - Well-known public marketplace: https://artifacthub.io/
- Running your own repository, e.g. via https://chartmuseum.com/

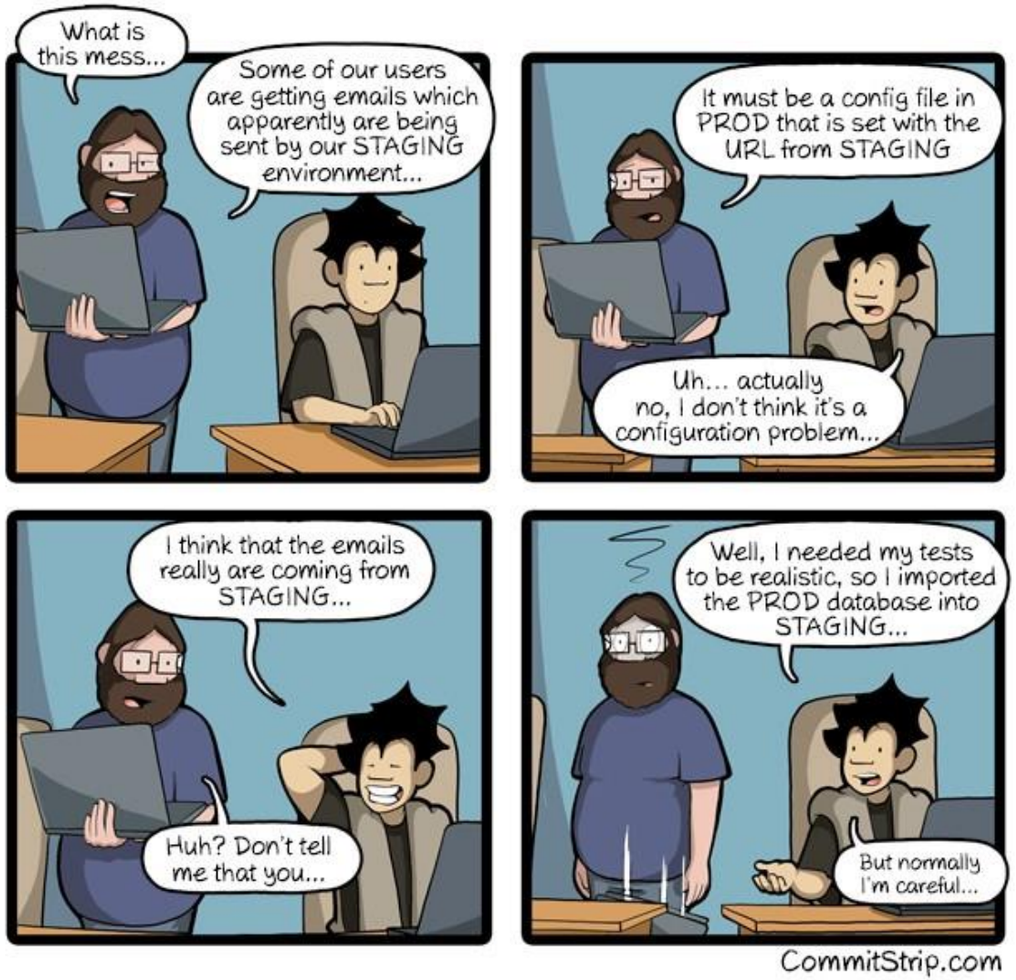
- Currently two packaging standards exist
  - Helm Repository (original and custom registry format)
  - OCI Registry (stored like container images, available since v3.8.0)
    - Less functionality with local Helm client (missing search, publish etc.)
- Usage
  - `$ helm repo add [NAME] [URL]  # adding 3rd-party repo`
  - `$ helm package [CHART_PATH]  # packaging your own chart`

Demo

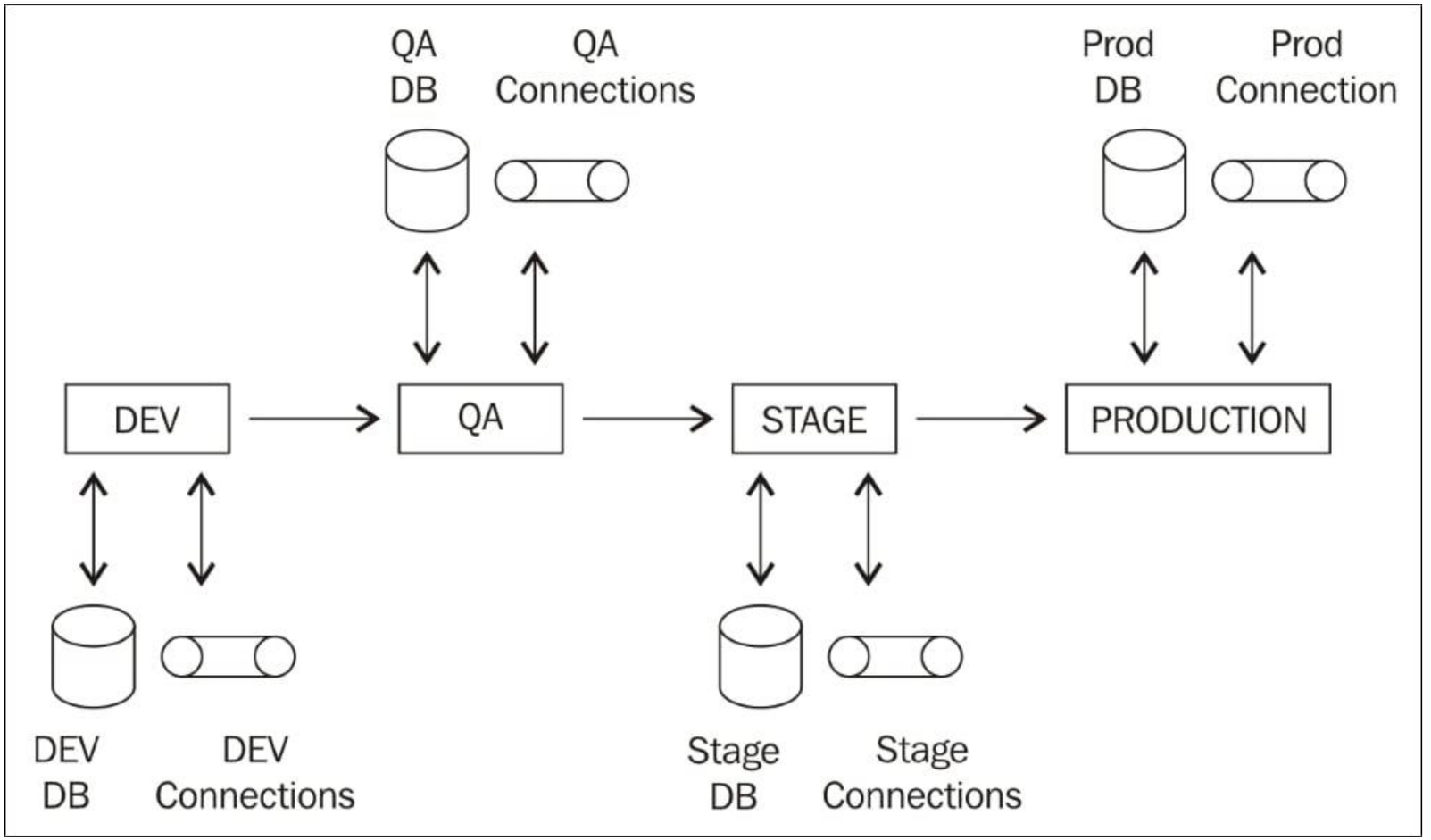
- **Start a new chart**
- **Installing a release**

- Very simple Helm chart can be found here:
  - https://github.com/cloudnativedevops/demo/tree/main/hello-helm3

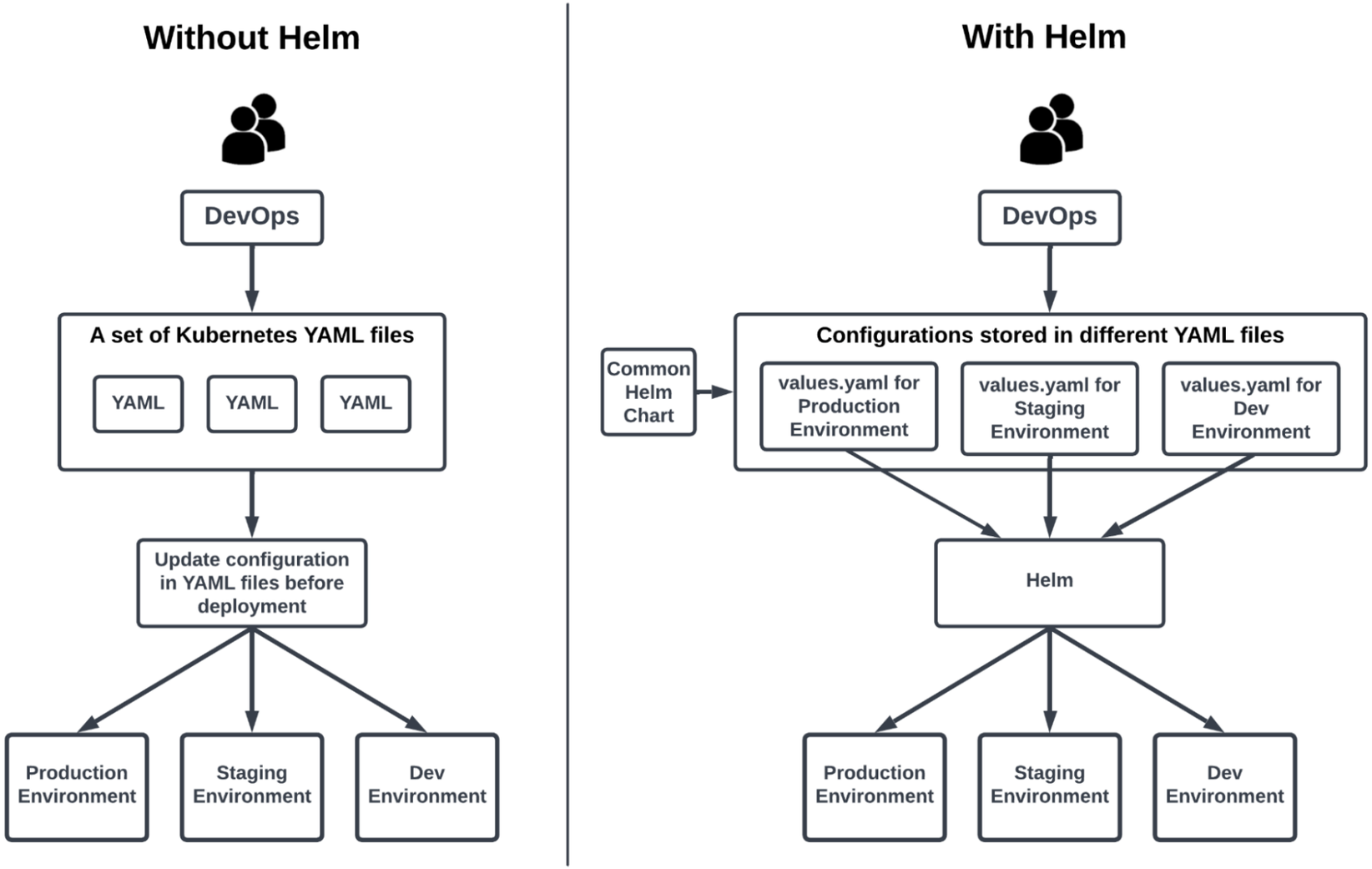
Stages in the Real World



Recap:
Environment configurations separate from application



How could Helm help with Staging?

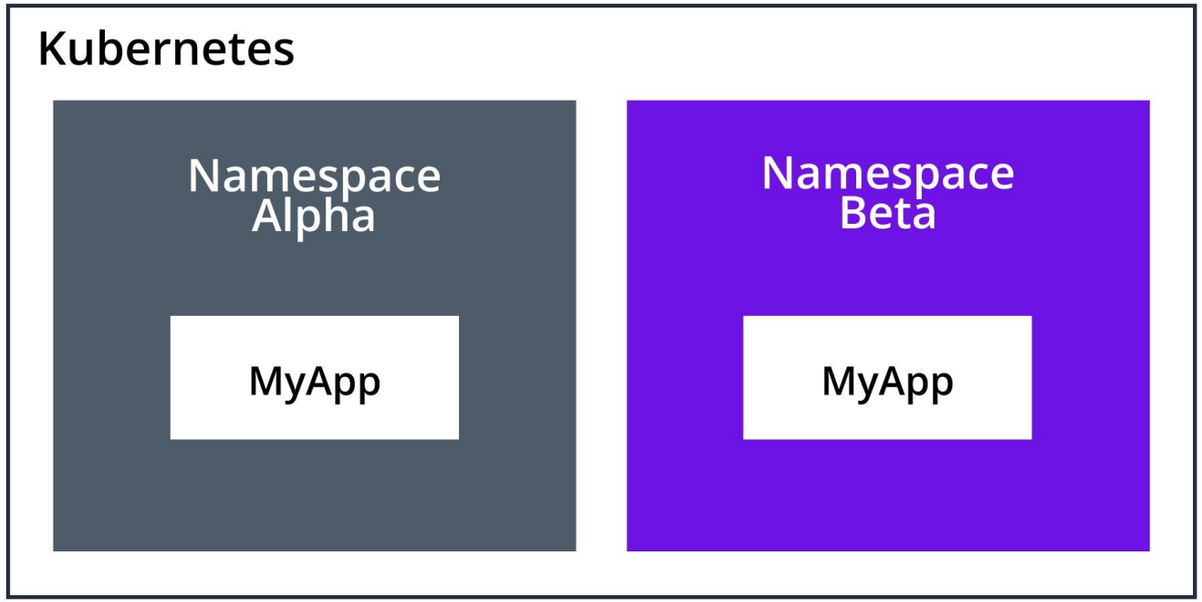


Source: https://circleci.com/blog/what-is-helm/

Staging and Namespaces

- Use the namespacing-concept for implementing different stages:
  - Stages have same network / infrastructure environment
  - Stages are accessible in a similar way

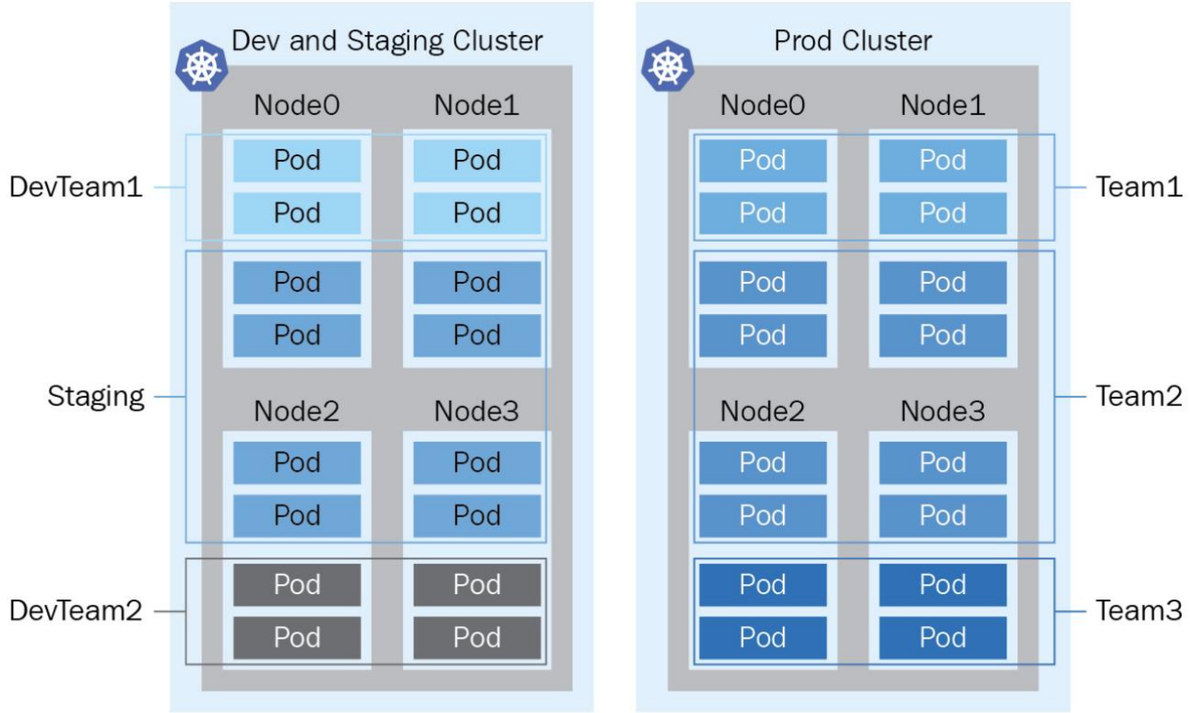
- Nevertheless:
  - Stages are isolated regarding the access, tokens, users, service accounts, resources.



Workloads are independent from infrastructure

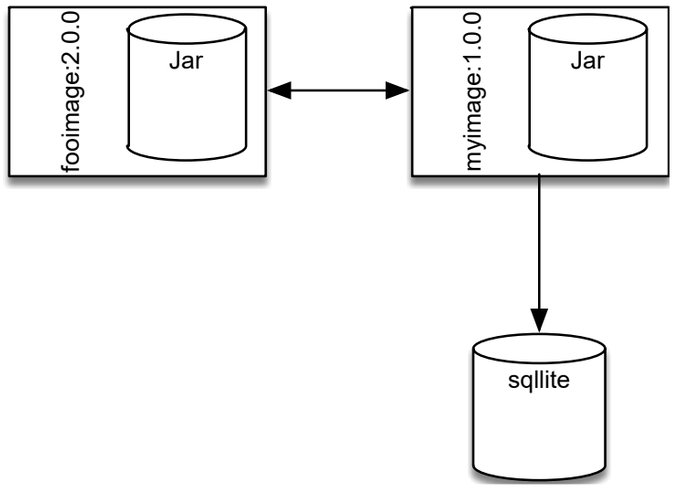
Pods might be scheduled independent of unterlaying infrastucture:

- Use different staging for clusters than for application and not:
  dev (App) → dev (K8s)
  int (App) → int (K8s)
  prod (App) → prod (K8s)
- Scheduling of pods to dedicated can be directed (however not in a very fine granular way)

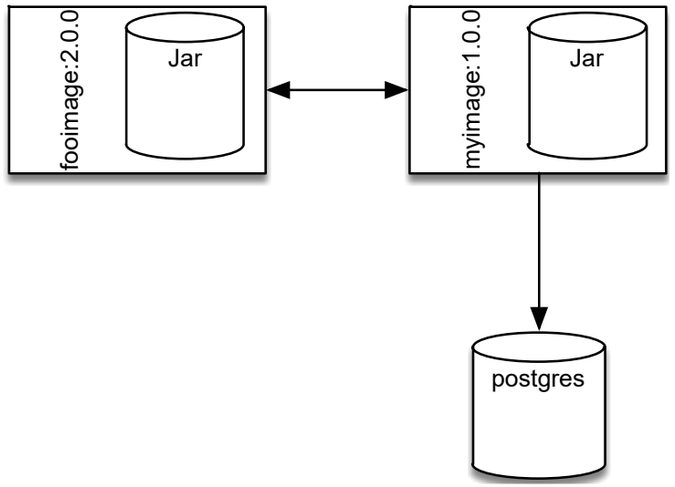


Dev/Prod Parity

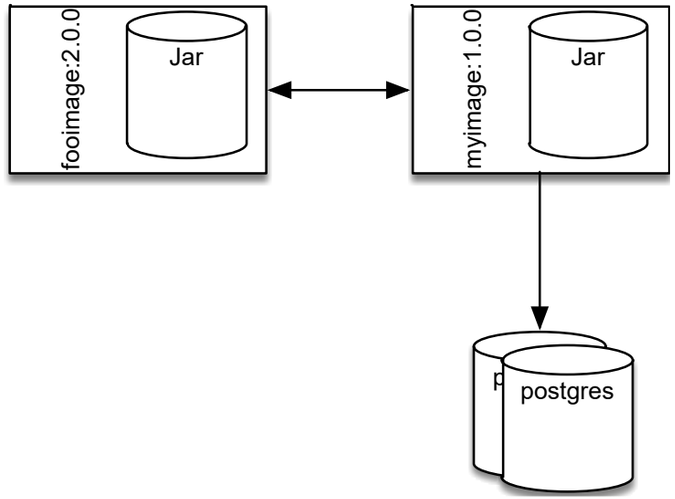
Dev



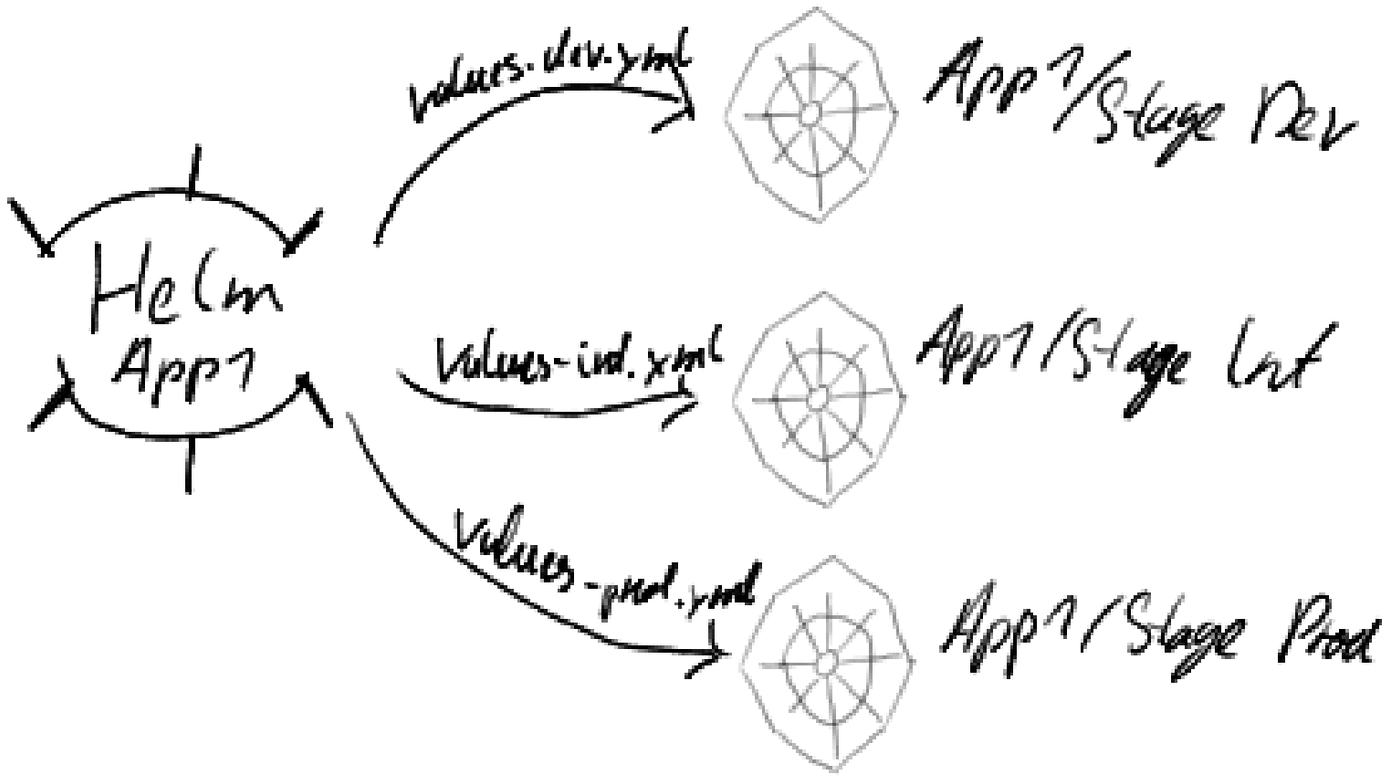
Integration



Production



Staging


Helm
App1
values-dev.yaml
values-int.yaml
values-prod.yaml
App1/Stage Dev
App1/Stage Int
App1/Stage Prod

Labels are important

Labels are first-class citizens in the cloud.

- All metainformation should be applied to labels/annotations.
  - Labels: identifying attributes to resources used for grouping, organizing
  - Annotations: non-identifying attributes of resources used for context to functionality or contact
- There are recommendations: https://kubernetes.io/docs/concepts/overview/working-with-objects/common-labels/
- Some labels are not to be modified
  - Labels to Deployment → partially immutable
  - Labels to Pod-Template → mutable

Labels within Prometheus on K8s

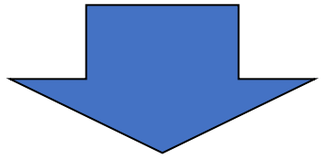
Use labels to distinguish between similar metrics

- **Metrics** should be common (latency, memory)
- **Labels** should offer filtering (endpoint, element)

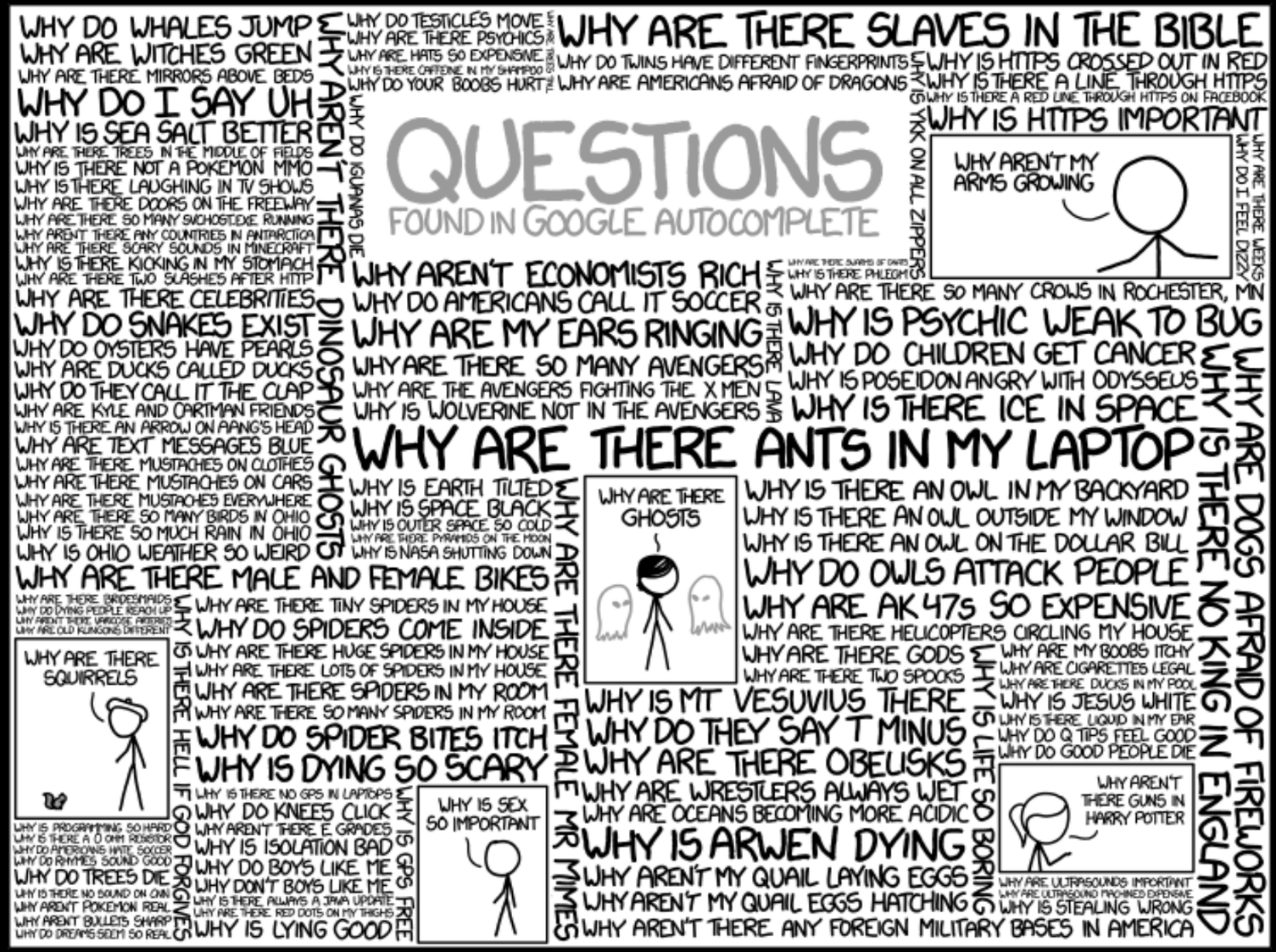
Different labels are applied to a metric

- Labels from the Pod/Service itself are mapped to the metrics of the Pod/Service
- Some Labels can be derived from K8s itself:
  - Kubernetes/pod/name
  - Kubernetes/namespace
- Some Labels are set by Prometheus itself
  - Job

```
http_requests_login_total
http_requests_logout_total
http_requests_adduser_total
http_requests_comment_total
http_requests_view_total
```



```
http_requests_total{path="/login"}
http_requests_total{path="/logout"}
http_requests_total{path="/adduser"}
http_requests_total{path="/comment"}
http_requests_total{path="/view"}
```


QUESTIONS
FOUND IN GOOGLE AUTOCOMPLETE
WHY DO WHALES JUMP
WHY ARE WITCHES GREEN
WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SYCHOSTICKS RUNNING
WHY AREN'T THERE ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY ARE THERE MUSTACHES ON CARS
WHY ARE THERE MUSTACHES EVERYWHERE
WHY ARE THERE SO MANY BIRDS IN OHIO
WHY IS THERE SO MUCH RAIN IN OHIO
WHY IS OHIO WEATHER SO WEIRD
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REARM UP
WHY AREN'T THERE VARICOSE AIRBIKES
WHY ARE OLD KLINGONS DIFFERENT
WHY ARE THERE SQUIRRELS
WHY IS PROGRAMMING SO HARD
WHY IS THERE A 0 OHM RESISTOR
WHY DO AMERICANS HATE SOCCER
WHY DO RHYMES SOUND GOOD
WHY DO TREES DIE
WHY IS THERE NO SOUND ON CNN
WHY AREN'T POKEMON REAL
WHY AREN'T BULLETS SHARP
WHY DO DREAMS SEEM SO REAL
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPIDER BITES ITCH
WHY IS DYING SO SCARY
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK
WHY AREN'T THERE E GRADES
WHY IS ISOLATION BAD
WHY DO BOYS LIKE ME
WHY DON'T BOYS LIKE ME
WHY IS THERE ALWAYS A JAVA UPDATE
WHY ARE THERE RED DOTS ON MY THIGHS
WHY IS LYING GOOD
WHY DO TESTICLES MOVE
WHY ARE THERE PSYCHICS
WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY DO YOUR BOOBS HURT
WHY ARE THERE SLAVES IN THE BIBLE
WHY DO TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE AMERICANS AFRAID OF DRAGONS
WHY IS HTTPS CROSSED OUT IN RED
WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON FACEBOOK
WHY IS HTTPS IMPORTANT
WHY AREN'T MY ARMS GROWING
WHY IS YKK ON ALL ZIPPERS
WHY ARE THERE WEEKS
WHY DO I FEEL DIZZY
WHY DO IGUANAS DIE
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY ARE THERE SO MANY AVENGERS
WHY ARE THE AVENGERS FIGHTING THE X MEN
WHY IS WOLVERINE NOT IN THE AVENGERS
WHY ARE THERE SWARMS OF GNATS
WHY IS THERE PHLEGM
WHY ARE THERE SO MANY CROWS IN ROCHESTER, MN
WHY IS PSYCHIC WEAK TO BUG
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY IS THERE LAVA
WHY ARE THERE ANTS IN MY LAPTOP
WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN
WHY ARE THERE GHOSTS
WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE AK 47s SO EXPENSIVE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOOKS
WHY IS MT VESUVIUS THERE
WHY DO THEY SAY T MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMING MORE ACIDIC
WHY IS ARWEN DYING
WHY AREN'T MY QUAIL LAYING EGGS
WHY AREN'T MY QUAIL EGGS HATCHING
WHY AREN'T THERE ANY FOREIGN MILITARY BASES IN AMERICA
WHY IS SEX SO IMPORTANT
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO GOOD PEOPLE DIE
WHY AREN'T THERE GUNS IN HARRY POTTER
WHY ARE ULTRASOUNDS IMPORTANT
WHY ARE ULTRASOUND MACHINES EXPENSIVE
WHY IS STEALING WRONG
WHY ARE DOGS AFRAID OF FIREWORKS
WHY IS THERE NO KING IN ENGLAND
WHY IS LIFE SO BORING
WHY IS GPS FREE
WHY ARE THERE FEMALE MR MIMES
WHY IS THERE HELL IF GOD FORGIVES
WHY AREN'T THERE DINOSAUR GHOSTS